

Annotated Audit of `Calculations.tex`

Shortcut-to-absorbing-target correction, with original-step replacements

31 May 2026

Contents

1	Provenance and how to read this version	1
2	Executive correction	2
3	What can be retained from the original	2
4	Critical step 1: the defect formula is not modular after absorption	2
5	Critical step 2: the first-passage correction has the wrong sign	3
6	Critical step 3: the auxiliary denominator and Fig. 1 geometry change	4
7	Critical step 4: replace the symmetric-case answer	4
8	Critical step 5: replace the beta-threshold conclusion	5
9	What this annotated version adds	6
10	Short referee-facing conclusion	6

1 Provenance and how to read this version

This is the referee-facing annotated version of Luca's original `Calculations.tex`. It is intentionally organised like a markup pass over the original manuscript: each critical original step is shown in red, followed by the correction in blue and the reason the replacement is necessary.

The original and audit files are archived beside this document:

Role	File	Notes
Original source	<code>source/original_from_outlook_Calculations.tex</code>	Outlook attachment, Luca, 28 May 2026
Original PDF	<code>source/original_from_outlook_Calculations.pdf</code>	Outlook attachment, Luca, 28 May 2026
Audit trail	<code>Calculations_marked_changes.v3.tex</code>	Issue-by-issue marked changes
Full correction	<code>Calculations_corrected_complete.v3.tex</code>	Expanded corrected derivation
This file	<code>luca_calculations_annotated_audit.tex</code>	Original-step correction map

Line anchors below refer to the archived original source. The main mathematical fault begins in the shortcut section around original lines 383–464.

2 Executive correction

For the stochastic model in which a fraction

$$\lambda = \beta(1 - q)$$

of the lazy self-loop at shortcut source u is redirected to the absorbing target v , the shortcut is an extra killing channel in the transient state space. Therefore the corrected resolvent is

$$\tilde{S}_{n_0}(n, z) = \tilde{W}_{n_0}(n, z) - \frac{z\lambda \tilde{W}_{n_0}(u, z) \tilde{W}_u(n, z)}{1 + z\lambda \tilde{W}_u(u, z)}.$$

The original denominator $1 - z\lambda \tilde{W}_u(u, z)$ and the original positive propagator correction have the wrong sign for a shortcut into an absorbing target.

The corrected first-passage generating function is

$$\tilde{F}_{n_0}(v, z) = \tilde{F}_{n_0}^{(0)}(v, z) + \frac{z\lambda \tilde{W}_{n_0}(u, z) [1 - \tilde{F}_u^{(0)}(v, z)]}{1 + z\lambda \tilde{W}_u(u, z)}. \quad (1)$$

This is positive relative to the no-shortcut hitting generating function for $0 \leq z < 1$, as a shortcut into the target must be.

3 What can be retained from the original

The absorbing-ring preliminaries are mostly usable after local corrections. In particular, the no-shortcut first-passage expansion and the killed-ring propagator strategy are the right starting point:

$$\mathcal{F}_{n_0}^{(0)}(m, t) = \sum_{\ell=1}^{N/2} f_{\ell}(n_0, m) \alpha_{\ell}^{t-1}, \quad \alpha_{\ell} = 1 - q + q \cos \frac{(2\ell - 1)\pi}{N},$$

with N even stated before using the $N/2$ odd-sector form.

What is wrong. The original absorbing-ring reduction contains a sign-bookkeeping problem in the double-sum intermediate line, and the subsequent initial-condition check attributes zero to individual sums that only cancel correctly in combination.

Replacement. Keep the final killed-ring propagator after spelling out the sign convention and the orthogonality sums. This is a local repair, not the main failure of the manuscript.

Added here. The full v3 derivation records the corrected finite-sum identities and the corrected orthogonality check; this annotated version uses those as the baseline and focuses on the shortcut section, where the conclusion changes.

4 Critical step 1: the defect formula is not modular after absorption

Original location: lines 383–399 of `Calculations.tex`.

The original manuscript first writes a probability-conserving one-defect formula and then replaces the free-ring propagator $\tilde{P}$ by the killed propagator $\tilde{W}$:

$$\tilde{S}_{n_0}(n, z) = \tilde{W}_{n_0}(n, z) + \frac{z\beta(1-q)\tilde{W}_{n_0}(u, z) [\tilde{W}_u(n, z) - \tilde{W}_v(n, z)]}{1 - z\beta(1-q) [\tilde{W}_u(u, z) - \tilde{W}_v(u, z)]}. \quad (\text{O1})$$

When v is the absorbing site, the original then simplifies this to

$$\tilde{S}_{n_0}(n, z) = \tilde{W}_{n_0}(n, z) + \frac{z\beta(1-q)\tilde{W}_{n_0}(u, z)\tilde{W}_u(n, z)}{1 - z\beta(1-q)\tilde{W}_u(u, z)}. \quad (\text{O2})$$

What is wrong. This replacement changes the model. In the probability-conserving ring, a shortcut transfers mass from u to another transient site. If the endpoint v is absorbing, the same transfer removes mass from the transient state space. The perturbation is therefore negative killing at u , not a positive rank-one return term.

Replacement. State the transient matrix first. If P_0 is the killed-ring transient matrix and $\lambda = \beta(1-q)$ is the shortcut-to-target probability, then

$$P = P_0 - \lambda|u\rangle\langle u|. \quad (2)$$

Therefore

$$\begin{aligned} (I - zP)^{-1} &= (I - zP_0 + z\lambda|u\rangle\langle u|)^{-1} \\ &= (I - zP_0)^{-1} - \frac{z\lambda(I - zP_0)^{-1}|u\rangle\langle u|(I - zP_0)^{-1}}{1 + z\lambda\langle u|(I - zP_0)^{-1}|u\rangle}. \end{aligned} \quad (3)$$

Taking the (n, n_0) element gives

$$\tilde{S}_{n_0}(n, z) = \tilde{W}_{n_0}(n, z) - \frac{z\lambda\tilde{W}_{n_0}(u, z)\tilde{W}_u(n, z)}{1 + z\lambda\tilde{W}_u(u, z)}. \quad (\text{C1})$$

Why. The sign is forced by probability monotonicity. Adding a shortcut into an absorbing target cannot increase transient occupation. The original formula does exactly that, so it cannot describe this stochastic model.

5 Critical step 2: the first-passage correction has the wrong sign

Original location: lines 401–414.

The original first-passage equation is

$$\tilde{F}_{n_0}(v, z) = \tilde{\mathcal{F}}_{n_0}(v, z) - \frac{z\beta(1-q)\tilde{W}_{n_0}(u, z) [1 - \tilde{\mathcal{F}}_u(v, z)]}{1 - z\beta(1-q)\tilde{W}_u(u, z)}. \quad (\text{O3})$$

What is wrong. This makes the shortcut contribution negative. For $0 \leq z < 1$, the quantities $\tilde{W}_{n_0}(u, z)$ and $1 - \tilde{F}_u^{(0)}(v, z)$ have non-negative coefficients. A target shortcut should increase, not decrease, the first-passage generating function.

Replacement. Summing the corrected transient propagator over nonabsorbing n and using

$$\tilde{R}_a^{(0)}(z) = \frac{1 - \tilde{F}_a^{(0)}(v, z)}{1 - z}$$

gives the corrected equation (1).

Added here. This is also the simplest diagnostic to show a reader where the original argument turns around: the sign of the shortcut contribution is visibly wrong before any Chebyshev algebra is used.

6 Critical step 3: the auxiliary denominator and Fig. 1 geometry change

Original location: lines 411–436.

The original defines

$$\tilde{H}_{N,|u-v|}(z) = \frac{\beta(1-q)}{q} \frac{1}{1 - z\beta(1-q)\tilde{W}_u(u, z)} \quad (\text{O4})$$

and in the symmetric case $|u - v| = N/2$ obtains

$$\tilde{H}_{N,N/2}(z) = \frac{1}{\frac{q}{\beta(1-q)} - \phi(z)}. \quad (\text{O5})$$

What is wrong. Because the denominator inherited the wrong sign, the pole picture based on a positive horizontal line intersecting $\phi(1/s)$ is the wrong picture for the stochastic shortcut-to-target model.

Replacement. With the corrected killing denominator, the same symbolic object becomes

$$\tilde{H}_{N,N/2}^{\text{corr}}(z) = \frac{1}{\frac{q}{\beta(1-q)} + \phi(z)}. \quad (\text{C2})$$

Equivalently, if one keeps the plotted curve $\phi(1/s)$, the horizontal line is $-q/[\beta(1-q)]$, not $+q/[\beta(1-q)]$.

Added here. The figure caption should also correct the typo “given the requirement $q > 1$ ”. The stochastic model requires $0 < q < 1$, and a nonzero shortcut probability proportional to $1 - q$ requires $q < 1$.

7 Critical step 4: replace the symmetric-case answer

For $N = 2L$, absorbing target $v = 0$, shortcut source $u = L$, and starting distance ρ from v , set

$$a = \frac{q}{\beta(1-q)}, \quad y = 1 + \frac{1}{q} \left(\frac{1}{z} - 1 \right).$$

The corrected closed form is

$$\tilde{F}_\rho(z) = \frac{aT_{L-\rho}(y) + U_{\rho-1}(y) + U_{L-\rho-1}(y)}{aT_L(y) + U_{L-1}(y)}, \quad U_{-1} = 0. \quad (4)$$

What is wrong. The original long convolution formula downstream of H is attached to the wrong denominator. It also contains several local issues: the baseline term uses $f_\ell(n_0, u)$ where the target term should be $f_\ell(n_0, v)$, and odd-sector terms involving $g_r^{(2)}$ are paired with γ_r in places where the odd-sector eigenvalue is α_r .

Replacement. Use (4) as the primary corrected result. Then obtain time-domain coefficients by residues at the roots of the corrected denominator

$$D(y) = aT_L(y) + U_{L-1}(y). \quad (5)$$

If y_j are its roots and $s_j = 1 - q + qy_j$, then

$$F_\rho(t) = \sum_{j=1}^L B_{\rho j} s_j^{t-1}, \quad B_{\rho j} = \frac{q N_\rho(y_j)}{D'(y_j)}, \quad (6)$$

where

$$N_\rho(y) = aT_{L-\rho}(y) + U_{\rho-1}(y) + U_{L-\rho-1}(y).$$

8 Critical step 5: replace the beta-threshold conclusion

Original location: line 464.

The original conclusion states that the second-peak condition is

$$\beta \leq \frac{2q}{(1-q)N}$$

and that the right tail changes between s_1^t and $t\alpha_1^t$ depending on this threshold.

What is wrong. That threshold comes from the wrong-sign pole geometry. The repeated pole that produces $t\alpha_1^t$ is an artefact of the uncorrected denominator, not a property of the corrected stochastic shortcut-to-target model.

Replacement. The corrected pole equation is

$$aT_L(y_s) + U_{L-1}(y_s) = 0, \quad y_s = 1 + \frac{s-1}{q}. \quad (7)$$

Writing $y_s = \cos \theta$ gives

$$\tan(L\theta) = -a \sin \theta. \quad (8)$$

The largest positive root satisfies

$$\gamma_1 < s_1 < \alpha_1, \quad \gamma_1 = 1 - q + q \cos \frac{2\pi}{N}, \quad \alpha_1 = 1 - q + q \cos \frac{\pi}{N}. \quad (9)$$

Thus the corrected right tail is

$$F_\rho(t) \sim B_{\rho 1} s_1^{t-1}, \quad t \rightarrow \infty, \quad (10)$$

not generically $t\alpha_1^t$.

Added here. The clean decay-dependence statement is the weak-shortcut expansion

$$s_1 = \alpha_1 - \frac{2\beta(1-q)}{N} + O([\beta(1-q)]^2), \quad (11)$$

so

$$1 - s_1 = 1 - \alpha_1 + \frac{2\beta(1-q)}{N} + O([\beta(1-q)]^2). \quad (12)$$

For large N ,

$$1 - s_1 \sim \frac{q\pi^2}{2N^2} + \frac{2\beta(1-q)}{N}. \quad (13)$$

9 What this annotated version adds

1. It pins down the stochastic interpretation before using a defect formula: the shortcut endpoint is absorbing, so the transient perturbation is killing.
2. It replaces the original sign pattern in the propagator and first-passage generating function.
3. It replaces the positive horizontal-line pole picture by the corrected $a + \phi = 0$ picture.
4. It gives the corrected symmetric generating function, pole equation, and tail expansion in one place.
5. It separates two questions that the original merges: the asymptotic right-tail decay and the finite-time visual existence of a second peak.

10 Short referee-facing conclusion

The absorbing-ring part of the original calculation can be repaired locally. The shortcut-to-target part cannot be repaired by small notation changes, because the original formula uses the probability-conserving shortcut defect after the shortcut endpoint has become absorbing. Under the stated stochastic model, the shortcut is an extra killing channel at u . This flips the sign of the rank-one correction and changes the denominator from

$$1 - z\beta(1 - q)\widetilde{W}_u(u, z) \quad \text{to} \quad 1 + z\beta(1 - q)\widetilde{W}_u(u, z).$$

Consequently the claimed threshold

$$\beta \leq \frac{2q}{(1 - q)N}$$

is not a valid analytical condition for the corrected model, and the corrected long-time tail is controlled by the simple pole s_1 solving $aT_L(y_s) + U_{L-1}(y_s) = 0$.